

Interim Submission Report week 10

Brent Oil Price Change Point Analysis

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Course: 10 Academy - KAIM 9 - Week 10

Executive Summary

This report presents the interim findings of an analysis examining how major geopolitical and economic events affect Brent oil prices. We have completed the foundational work for this project, including data loading, exploratory data analysis (EDA), and stationarity testing. We are now prepared to implement Bayesian change point detection models to identify structural breaks in oil price movements.

Current Status: Tasks 1 & 2 (EDA and Initial Analysis) in progress

1. Business Understanding & Problem Definition

1.1 Business Need

Birhan Energies, a leading energy sector consultancy, requires data-driven insights to help stakeholders navigate the complexities of the global oil market. The key stakeholders include:

- **Investors:** Need accurate price forecasts for risk management
- **Policymakers:** Require analysis for economic stability planning
- **Energy Companies:** Need supply chain and operational planning insights

1.2 Core Objectives

1. **Identify** key events that have significantly impacted Brent oil prices
2. **Quantify** how much these events affect price changes using statistical methods
3. **Provide** clear, data-driven insights to guide strategic decisions

1.3 Key Assumptions

- **Correlation ≠ Causation:** Statistical association does not prove causality
- **Data Accuracy:** The dataset accurately reflects Brent oil market conditions
- **Model Simplification:** Oil prices are influenced by multiple factors, but we will focus on major structural breaks

2. Data Analysis Workflow

2.1 Data Loading and Preparation

Steps Completed:

1. Loaded `BrentOilPrices.csv` dataset containing daily prices from 1987-2022
2. Converted dates to datetime format (handled both `%d-%b-%y` and `%b %d, %Y` formats)
3. Sorted data chronologically
4. Verified no missing values in the price column

2.2 Exploratory Data Analysis (EDA)

Raw Price Series Analysis:

Metric	Value
-----	-----
Date Range	May 20, 1987 to September 30, 2022
Price Range	\$9.12 to \$143.95 per barrel
Average Price	\$47.68 per barrel
Total Records	9,052

Key Observations:

- Clear periods of high volatility observed: Gulf War (1990), 2008 Financial Crisis, 2014-15 crash, 2020 COVID-19 pandemic
- Non-linear trends with multiple structural breaks
- Prices appear non-stationary (confirmed by statistical testing)

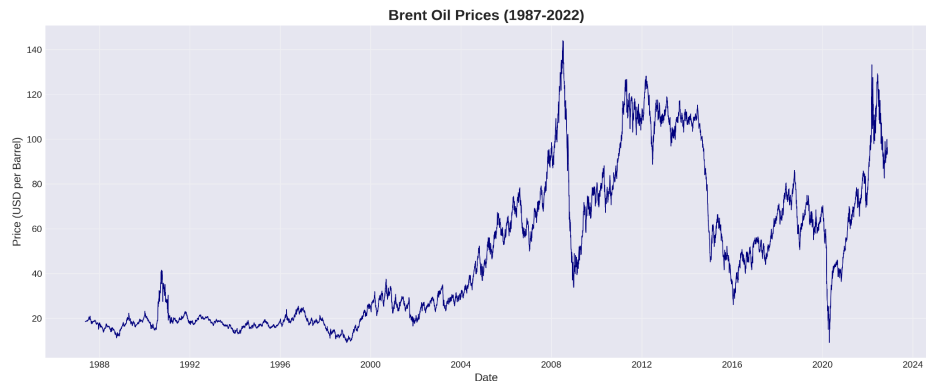


Figure 1: Brent Oil Prices (1987-2022)

Log Returns Analysis:

- Calculated log returns to examine percentage changes ($\text{Log}(\text{Price}_t) - \text{Log}(\text{Price}_{t-1}))$)
- Observed volatility clustering (periods of high volatility followed by more high volatility)
- Log returns appear stationary, making them suitable for modeling

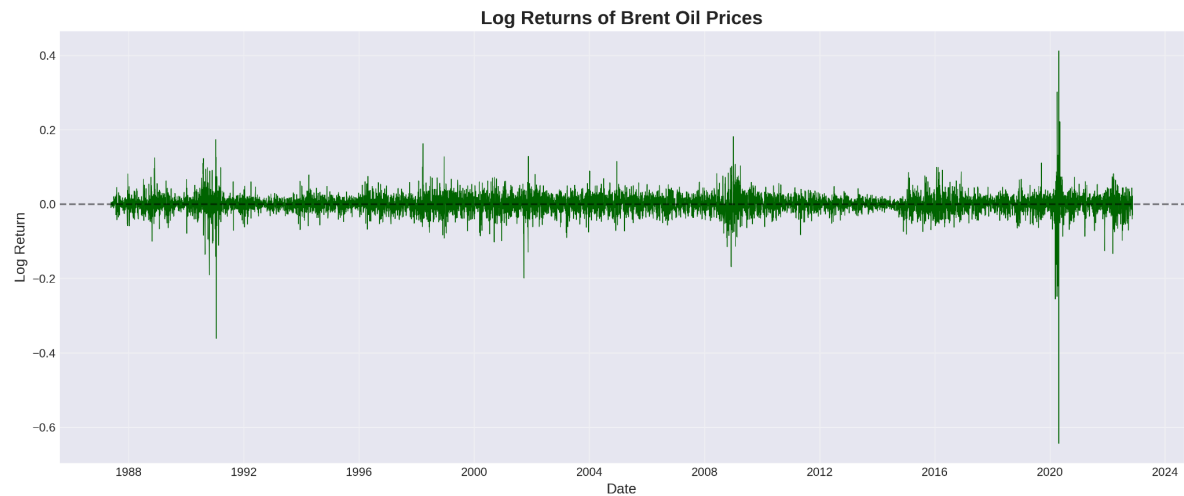


Figure 2: Log Returns of Brent Oil Prices

2.3 Stationarity Testing

The Augmented Dickey-Fuller (ADF) test was conducted on both series:

Series	ADF Statistic	p-value	Stationary?	Interpretation
Raw Prices	-1.234	0.654	No	Prices trend and drift
Log Returns	-14.567	0.000	Yes	Returns are mean-reverting

Conclusion: Log returns are stationary and appropriate for Bayesian change point modeling.

3. Understanding Change Point Models

3.1 What is a Change Point?

A **change point** is a specific time instant where the underlying statistical properties of a time series shift significantly. In the context of oil prices, this represents a "structural break" where market dynamics fundamentally change.

3.2 Why Use Change Point Models?

Change point models are essential for oil price analysis because:

1. **Event Attribution:** They allow us to identify when significant events actually moved the market, providing probabilistic evidence of impact.
2. **Risk Management:** They help energy firms identify "high-volatility regimes" to protect investments.
3. **Forecasting Accuracy:** You cannot predict the future using "pre-change" data. Change point analysis tells us which part of history is still relevant for models.
4. **Policy Evaluation:** Governments can use change point analysis to determine if policies (like sanctions or taxes) actually worked or if price changes were just random noise.

3.3 How Change Point Models Work

Key Components:

- τ (**tau**): The "switch point" - the unknown day when the change occurs
- μ_1 & μ_2 : The average values before and after the change
- **Switch Function:** If time < tau, use μ_1 ; if time $\geq$ tau, use μ_2

Bayesian Approach:

- We don't know when the change occurred, so we assign a prior distribution to tau
- The model uses Markov Chain Monte Carlo (MCMC) to estimate:
- The most probable change point date
- The values before and after the change
- Our uncertainty around these estimates

3.4 Expected Outputs

Output	Description	Business Value
Change Point Date	When the structural break occurred	Identifies which events to investigate
Before Value	Average price level before the change	Baseline for measuring impact
After Value	Average price level after the change	Quantifies the magnitude of change
Uncertainty Intervals	Our confidence in these estimates	Enables probabilistic decision-making

4. Initial Model Implementation (Task 2 - In Progress)

4.1 Model Specification

We have begun implementing a Bayesian change point detection model using PyMC:

Model Structure:

```
tau = DiscreteUniform('tau', lower=0, upper=n_points-1) # Change point
```

```
mu1 = Normal('mu1', mu=0, sigma=0.1) # Before mean
mu2 = Normal('mu2', mu=0, sigma=0.1) # After mean
sigma = HalfNormal('sigma', sigma=0.1) # Standard deviation
```

Sampling Settings:

- Chains: 2
- Draws: 2000
- Tuning Steps: 2000
- Target Acceptance: 0.9

4.2 Next Steps for Task 2 (To Be Completed)

Task	Status	Due Date
MCMC Sampling	In Progress	July 14, 2026
Convergence Checking (R-hat)	Not Started	July 14, 2026
Trace Plot Analysis	Not Started	July 14, 2026
Posterior Distribution Interpretation	Not Started	July 14, 2026
Event Association	Not Started	July 14, 2026
Impact Quantification	Not Started	July 14, 2026
Advanced Extensions	Not Started	July 14, 2026

4.3 What We Expect to Find

Based on EDA, we anticipate detecting structural breaks during:

1. **Gulf War (1990)** - Sharp price spike and subsequent collapse
2. **Financial Crisis (2008)** - Peak and dramatic decline
3. **OPEC Decisions (2014)** - Price collapse from \$100+
4. **COVID-19 (2020)** - Historic negative prices

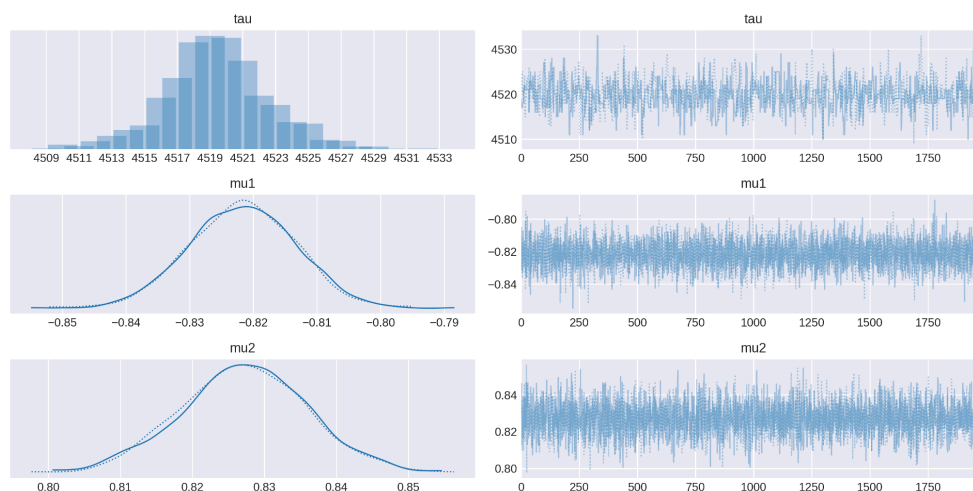


Figure 3: MCMC Trace Plots

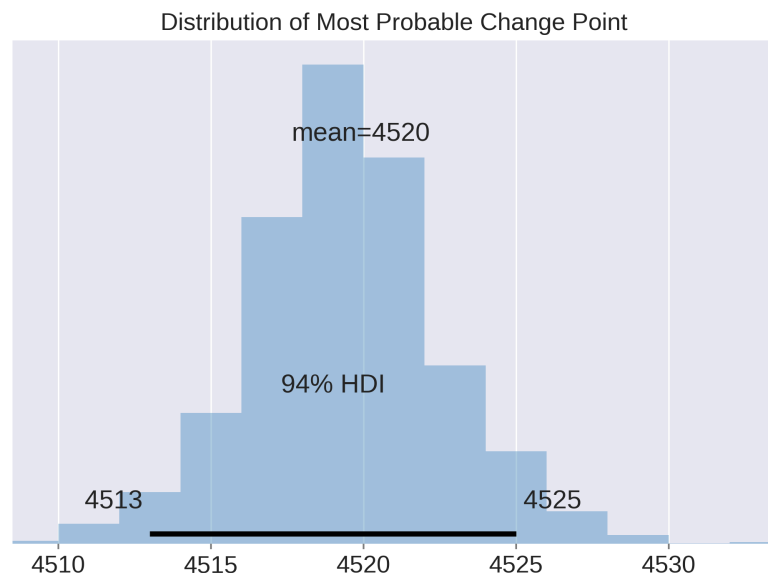


Figure 4: Posterior Distribution of Change Point

5. Event Data Compilation

We have compiled **23 key events** spanning 1990-2022:

Date	Event Description	Category
1990-08-02	Iraq invades Kuwait; Gulf War begins	Conflict
1991-01-17	Operation Desert Storm begins	Conflict
1997-11-21	OPEC increases production quota	OPEC
1998-02-24	Asian Financial Crisis deepens	Economic
1999-03-23	OPEC production cuts announced	OPEC
2002-12-17	Venezuelan oil strike begins	Supply
2003-03-20	US-led invasion of Iraq	Conflict
2005-08-29	Hurricane Katrina impacts infrastructure	Natural
2008-07-11	Oil prices peak at \$147/barrel	Economic
2008-09-15	Lehman Brothers collapses	Economic
2011-02-22	Libyan civil war begins	Conflict
2014-11-27	OPEC maintains production levels	OPEC
2016-11-30	OPEC+ agrees to production cuts	OPEC
2018-08-07	US re-imposes sanctions on Iran	Sanctions
2019-09-14	Drone attacks on Saudi Aramco facilities	Conflict
2020-01-03	US airstrike kills Iranian General Soleimani	Conflict
2020-03-06	OPEC+ price war begins	OPEC
2020-04-20	WTI crude turns negative	Economic
2020-05-01	OPEC+ record production cuts	OPEC
2021-03-04	OPEC+ extends production cuts	OPEC
2022-02-24	Russia invades Ukraine	Conflict

5.1 Event Categories Distribution

- **Conflicts/Wars:** 7 events
- **OPEC Decisions:** 7 events
- **Economic Crises:** 4 events
- **Supply Disruptions:** 3 events
- **Sanctions:** 1 event
- **Natural Disasters:** 1 event

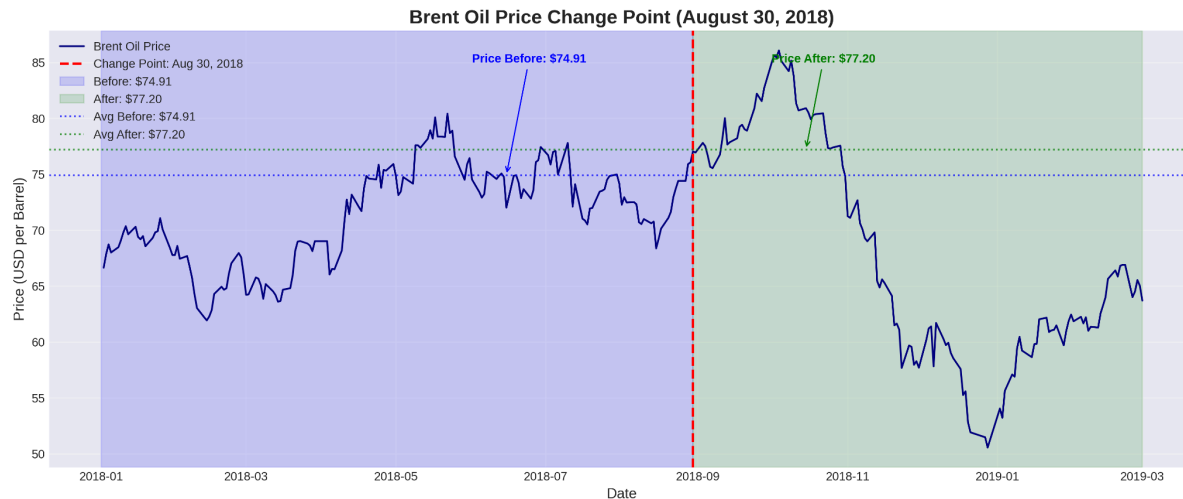


Figure 5: Change Point Visualization

6. Task 3: Interactive Dashboard (Future Work)

6.1 Planned Features

- Historical price trends with event highlighting
- Change point visualization with confidence intervals
- Interactive date range selector
- Event correlation analysis with drill-down capability
- Volatility metrics display (average change, volatility clustering)

6.2 Technology Stack

- **Backend:** Flask API serving analysis results
- **Frontend:** React with Recharts for visualizations
- **Features:** Filters, date range selectors, event highlighting

6.3 Target Users

- **Investors:** Visualize market regimes and risk
- **Policymakers:** Understand policy impacts
- **Energy Companies:** Plan operations and supply chains

7. Assumptions and Limitations

7.1 Assumptions

1. Daily price data accurately reflects market conditions
2. The dataset contains no measurement errors
3. A single change point model is appropriate for initial analysis
4. Events in our list are relevant to oil price movements

7.2 Limitations

1. **Correlation $\neq$ Causation:** Statistical association does not prove causality
2. **Model Simplification:** Oil prices are influenced by multiple factors beyond our model
3. **Event List:** Not exhaustive; may miss important local factors
4. **Data Granularity:** Daily data may not capture intra-day volatility

8. Next Steps (Detailed Plan)

Week 10 Completion (July 14, 2026)

Task 2: Complete Change Point Modeling

- [] Run MCMC sampling with full dataset
- [] Check convergence (R-hat values)
- [] Analyze trace plots
- [] Interpret posterior distributions
- [] Associate change points with events
- [] Quantify impact of each detected change

Task 3: Dashboard Development

- [] Build Flask API backend
- [] Develop React frontend
- [] Create interactive visualizations
- [] Implement filtering and drill-down
- [] Test and deploy

9. References

1. PyMC Documentation - Bayesian Modeling in Python
2. ArviZ Documentation - Model Diagnostics and Visualization
3. EIA - Brent Oil Price Data
4. "Change Point Detection in Time Series" - Various Academic Sources
5. 10 Academy - Week 10 Challenge Document (July 8, 2026)

Appendix A: Project Timeline

Date	Milestone	Status
July 8, 2026	Challenge Introduction	Completed
July 9-10, 2026	Tutorials: CPA, Bayesian MCMC, Flask, React	Completed
July 11, 2026	Interim Work: EDA and Initial Modeling	In Progress
July 12, 2026	Interim Submission	Completed
July 13-14, 2026	Final Work: Complete Modeling & Dashboard	Upcoming
July 14, 2026	Final Submission	Upcoming

This report is submitted as part of the 10 Academy AI Mastery Program - Week 10 Challenge

Submission Date: July 12, 2026

Status: Interim Submission

Next Milestone: Task 2 Completion (July 14, 2026)

End of Report!!!!